

Cuemath AI Tutor Screener — Final Workplan

100% Free | 2 API Keys Only | Simple Stack

Tech Stack (Final)

Layer	Tool	Cost
Frontend	Next.js 14 + Tailwind CSS + Three.js	Free
Voice Input (STT)	Groq Whisper API	Free forever
Voice Output (TTS)	Browser SpeechSynthesis API	Free built-in
AI Conversation	Anthropic Claude claude-sonnet-4-20250514	\$5 free credits on signup
Backend	FastAPI (single main.py file)	Free
Database	SQLite (built into Python, zero setup)	Free
Frontend Deploy	Vercel	Free tier
Backend Deploy	Railway	Free tier

API keys needed: 2

- `GROQ_API_KEY` — signup at console.groq.com (free forever)
- `ANTHROPIC_API_KEY` — signup at console.anthropic.com (\$5 free credits)

Project Structure

```
cuemath-screener/  
├─ frontend/  
│   ├─ app/  
│   │   ├─ page.tsx                ← welcome screen  
│   │   ├─ interview/page.tsx      ← main interview screen  
│   │   └─ report/[id]/page.tsx    ← recruiter report screen  
│   └─ components/  
│       ├─ AudioOrb.tsx            ← Three.js orb  
│       └─ ScoreCard.tsx          ← report dimension cards  
│   └─ hooks/  
│       └─ useInterview.ts         ← state machine + voice loop  
└─ backend/  
    ├─ main.py                    ← entire backend (5 endpoints, ~180  
lines)  
    ├─ prompts.py                ← Claude system prompt + scoring prompt  
    ├─ db.py                     ← SQLite setup  
    └─ requirements.txt
```

Backend (main.py) — 5 Endpoints Only

```
python
```

```

from fastapi import FastAPI, UploadFile
from fastapi.middleware.cors import CORSMiddleware
import anthropic, groq, sqlite3, uuid, json, base64
from prompts import ARIA_SYSTEM_PROMPT, SCORING_PROMPT

app = FastAPI()
app.add_middleware(CORSMiddleware, allow_origins=["*"])

groq_client = groq.Groq(api_key=GROQ_API_KEY)
claude_client = anthropic.Anthropic(api_key=CLAUDE_API_KEY)

# 1. Start session
@app.post("/interview/start")
def start():
    sid = str(uuid.uuid4())
    db_exec("INSERT INTO sessions VALUES (?,?)", [sid, "active"])
    return {"session_id": sid}

# 2. Transcribe audio (Groq Whisper)
@app.post("/interview/transcribe")
async def transcribe(audio: UploadFile):
    result = groq_client.audio.transcriptions.create(
        model="whisper-large-v3",
        file=("audio.webm", await audio.read()),
        language="en"
    )
    return {"transcript": result.text}

# 3. Get Aria's reply (Claude)
@app.post("/interview/respond")
def respond(payload: dict):
    # payload: { history: [...messages], current_question: str }

```

```

res = claude_client.messages.create(
    model="claude-sonnet-4-20250514",
    max_tokens=250,
    system=ARIA_SYSTEM_PROMPT.replace("{QUESTION}", payload["current_questi
    messages=payload["history"]
)
text = res.content[0].text
# strip intent tag before returning spoken text
spoken = text.split("[INTENT:")[0].strip()
intent = "SUFFICIENT"
if "[INTENT:" in text:
    intent = text.split("[INTENT:")[1].replace('"', "").strip()
return {"text": spoken, "intent": intent}

# 4. End interview + score
@app.post("/interview/end")
def end(payload: dict):
    # payload: { session_id: str, history: [...] }
    transcript = "\n\n".join(
        f"'{CANDIDATE}' if m['role']=='user' else 'ARIA': {m['content']}"
        for m in payload["history"]
    )
    res = claude_client.messages.create(
        model="claude-sonnet-4-20250514",
        max_tokens=1500,
        messages=[{"role": "user", "content": SCORING_PROMPT.replace("{TRANSCRIPT
    )
    report = json.loads(res.content[0].text)
    rid = str(uuid.uuid4())
    db_exec("INSERT INTO reports VALUES (?, ?, ?, ?)",
            [rid, payload["session_id"], transcript, json.dumps(report)])
    return {"report_id": rid, "report": report}

```

```
# 5. Fetch report
@app.get("/report/{rid}")
def get_report(rid: str):
    row = db_fetch("SELECT scores FROM reports WHERE id=?", [rid])
    return json.loads(row[0])
```

Prompts (prompts.py)

Aria System Prompt

```
python
```

```
ARIA_SYSTEM_PROMPT = """
```

```
You are Aria, a warm and professional interviewer for Cuemath, a children's mat  
You are screening tutor candidates for soft skills only – NOT math knowledge.  
You assess: communication clarity, patience, warmth, simplification ability, En
```

```
Rules:
```

- Keep all responses under 3 sentences
- Sound natural and warm, never robotic
- Always end with a question or prompt
- After each candidate response, decide:
 - SUFFICIENT → candidate gave a clear, specific answer – move on
 - PROBE_NEEDED → answer was vague or too short – ask one follow-up
 - REDIRECT → candidate went off-topic – bridge back gently
 - WRAP → all questions done – close the interview warmly
- Always append [INTENT:XXXX] at the very end of your message (candidate won't

```
Current question to ask or follow up on:
```

```
{QUESTION}
```

```
"""
```



Scoring Prompt

```
python
```

```
SCORING_PROMPT = """
```

```
Evaluate this tutor candidate interview for Cuemath.
```

```
Return ONLY valid JSON. No markdown, no explanation, no code fences.
```

```
{
  "verdict": "ADVANCE" or "REVIEW" or "DECLINE",
  "overall_score": <1-10 number>,
  "summary": "<2 sentence summary>",
  "recommendation": "<1 sentence for recruiter>",
  "dimensions": {
    "communication_clarity": {"score": <1-10>, "evidence": ["quote1", "quote2"]
    "patience_empathy":      {"score": <1-10>, "evidence": ["quote1", "quote2"]
    "simplification_ability": {"score": <1-10>, "evidence": ["quote1", "quote2"]
    "english_fluency":        {"score": <1-10>, "evidence": ["quote1", "quote2"]
    "warmth_enthusiasm":      {"score": <1-10>, "evidence": ["quote1", "quote2"]
  }
}
```

```
Verdict rules:
```

- ADVANCE = overall_score >= 7 AND no dimension below 5
- REVIEW = overall_score 5.5-6.9 OR one dimension below 5
- DECLINE = overall_score < 5.5 OR two+ dimensions below 5

```
TRANSCRIPT:
```

```
{TRANSCRIPT}
```

```
"""
```

Frontend — Interview Screen

Voice Loop (useInterview.ts)

START

- MediaRecorder starts (WebM/Opus)
- Silence VAD: if mic quiet for 1.5s → stop recording
- POST audio blob → /interview/transcribe → get transcript
- Show candidate transcript on screen
- POST history → /interview/respond → get { text, intent }
- Show Aria text as subtitle
- Browser SpeechSynthesis speaks Aria's text
- On speech end → check intent:
 - PROBE_NEEDED / REDIRECT → loop back to MediaRecorder
 - SUFFICIENT → advance question index → loop
 - WRAP → POST /interview/end → go to report page

Silence VAD (simple, no library)

typescript

```
// Web Audio API AnalyserNode approach
const analyser = audioCtx.createAnalyser();
const data = new Uint8Array(analyser.frequencyBinCount);

function checkSilence() {
  analyser.getByteFrequencyData(data);
  const avg = data.reduce((a, b) => a + b) / data.length;
  if (avg < 8) silenceFrames++; // threshold: very quiet
  else silenceFrames = 0;
  if (silenceFrames > 45) stopRecording(); // 1.5s at 30fps
  requestAnimationFrame(checkSilence);
}
```

Three.js Audio Orb (AudioOrb.tsx)

typescript

```
// Sphere with 64x64 segments
// In animation loop: displace vertices using sin waves + analyser amplitude
// orbState prop controls color:
//   'listening' → #3b82f6 (blue)
//   'thinking'  → #ffffff at 30% opacity (slow pulse)
//   'speaking'  → #f59e0b (amber)
//   'idle'      → #6b7280 (gray)

// Vertex displacement:
positions[i]      += Math.sin(time + i * 0.5) * amplitude * 0.3;
positions[i + 1] += Math.cos(time + i * 0.3) * amplitude * 0.3;
```

SpeechSynthesis (TTS — free)

typescript

```
function ariaSpeak(text: string, onEnd: () => void) {  
  const utter = new SpeechSynthesisUtterance(text);  
  utter.rate = 0.90; // slightly slower = warmer feel  
  utter.pitch = 1.10;  
  // pick best female voice available  
  const voices = speechSynthesis.getVoices();  
  utter.voice = voices.find(v =>  
    v.name.includes('Samantha') || // macOS  
    v.name.includes('Google UK') || // Chrome  
    v.name.includes('Female')  
  ) || voices[0];  
  utter.onend = onEnd;  
  speechSynthesis.speak(utter);  
}
```

Screens

1. Welcome Screen (page.tsx)

- Cuemath logo
- "Hi! I'm Aria, your interviewer today."
- Name input field
- "Allow microphone" button (triggers getUserMedia)
- Mic test: candidate speaks, sees waveform confirm it works
- "Start Interview" CTA

2. Interview Screen (interview/page.tsx)

- Full dark background (● #0f0f1a)
- Center: Three.js orb (color changes with state)
- Bottom center: Aria's current text (subtitle style, white)
- Top center: Candidate's transcript (muted gray, smaller)
- No timer, no score, no rubric visible to candidate
- Small "End early" button bottom-right (emergency only)

3. Thank You Screen

- "Thank you, [Name]! We'll review your session and be in touch within 24 hours."
- Cuemath branding
- No score shown to candidate

4. Report Screen (report/[id]/page.tsx)

- Verdict badge: ADVANCE (green) / REVIEW (amber) / DECLINE (red)
- Overall score: large number center
- 5 dimension cards:
 - Dimension name
 - Score / 10
 - Animated fill bar (Framer Motion)
 - 2 evidence quotes in italics
- Summary paragraph
- Recruiter recommendation

- Full transcript (collapsed by default, expand button)
 - "Save as PDF" → browser print dialog
-

Question Bank (5 Questions)

typescript

```
const QUESTIONS = [  
  // Q1 - simplification + clarity  
  "Let's start - can you explain what a fraction is to a 9-year-old who has nev  
  
  // Q2 - patience + empathy  
  "A student has been staring at a problem for 5 minutes and says 'I just don't  
  
  // Q3 - warmth + handling frustration  
  "How do you handle a student who keeps getting wrong answers and is starting  
  
  // Q4 - real example probe  
  "Tell me about a time you had to explain something complicated in a simple wa  
  
  // Q5 - professionalism + warmth  
  "A parent messages saying their child hasn't improved in 3 weeks. How do you  
];
```



Edge Cases

Situation	Handler
Vague / short answer	Claude returns PROBE_NEEDED → follow-up (max 2x per question)
Long tangent	Claude returns REDIRECT → bridge back
8s silence after question	Aria says "Take your time, I'm listening" via SpeechSynthesis
Mic denied	Show setup guide: "Please allow mic access in your browser settings"
Groq API error	Retry once → if fails again, Aria says "Could you repeat that?"
Non-English detected	Claude notes in fluency score automatically
SpeechSynthesis not loading voices	Add 500ms delay + getVoices() retry (browser quirk fix)

Sprint Plan (5 Days)

Day 1 — Backend

- ☐ FastAPI project setup
- ☐ SQLite db.py (2 tables: sessions, reports)
- ☐ All 5 endpoints working
- ☐ Test with Postman: transcribe audio file → get transcript

Day 2 — Voice loop

- ☐ Next.js app setup + Tailwind
- ☐ MediaRecorder → POST to /transcribe → show transcript
- ☐ Silence VAD working
- ☐ POST to /respond → SpeechSynthesis speaks reply
- ☐ Full loop working (ugly UI, just functional)

Day 3 — Interview UI

- ☐ Three.js AudioOrb with 3 color states
- ☐ Interview screen layout (dark, centered)
- ☐ State machine: IDLE → LISTENING → THINKING → SPEAKING
- ☐ Framer Motion transitions between states

Day 4 — Scoring + Report

- ☐ /interview/end endpoint + Claude scoring
- ☐ Report page: verdict badge, score bars, evidence quotes
- ☐ Framer Motion reveal animations on report

Day 5 — Polish + Deploy

- ☐ Welcome screen + thank you screen
 - ☐ All edge case handlers
 - ☐ Deploy: Vercel (frontend) + Railway (backend)
 - ☐ Full test: complete 10-min interview → report generated
-

Total Cost

Groq Whisper	\$0 (free forever)
Anthropic Claude	\$0 (\$5 free credits, ~100 interviews)
Railway backend	\$0 (free tier)
Vercel frontend	\$0 (free tier)
Total	\$0

API Keys Needed

1. `GROQ_API_KEY` → console.groq.com → Create API Key (free, no card)
2. `ANTHROPIC_API_KEY` → console.anthropic.com → Get API Key (\$5 free credits)